

Project Overview

This project aims to modernize a small library by creating a cohesive database system for the inventory and memberships. The database will store all items in the library, track transactions by clients, generate reports, generate fines, and maintain a record of other relevant information.

Scope

On the staff side, the system should be able to handle storage of a variety of media with associated information, track borrowing and return transactions, and also generate reports of trends, fines, and client activity. On the client side, the system will track ID, name, and contact information, and enforce membership fees/limits/etc.

Glossary

1. SSN - Social Security Number
2. ISBN - International Standard Book Number
3. ISSN - International Standard Serial Number
4. DOB - Date of Birth

Relation Schema

Staff

Attribute Name	Data Type	Constraints	Description
Staff ID	INT(12)	PK , NOT NULL	Staff member ID
Name	VARCHAR(20)	NOT NULL	Staff member name
Age	INT(3)	NOT NULL	Staff member age
Phone Number	INT(10)	NOT NULL	Staff member phone #
Address	VARCHAR(20)	NOT NULL	Staff member address
SSN	CHAR(13)	NOT NULL	Staff member social security number
DOB	DATE	NOT NULL	Staff member date of birth

Client

Attribute Name	Data Type	Constraints	Description
ClientID	INT or VARCHAR(...)	PK , NOT NULL,	Unique identifier for each

		UNIQUE	client
Name	VARCHAR(50)	NOT NULL	Client's full
Contact Information	VARCHAR(100)	May be NULL	Phone/email/etc.
Membership	VARCHAR(20)	May be NULL	Name or label of the membership tier
Account	VARCHAR(20)	May be NULL	Current status of the client's
MembershipTypeID	INT or VARCHAR(...)	FK→membership	Links to

Copy

Attribute Name	Data Type	Constraints	Description
Copy ID	INT(12)	PK , NOT NULL	ID of specific copy
Type	VARCHAR(20)	NOT NULL	Type of media
Availability Status	BIT	NOT NULL	If the copy is available to be borrowed
ID	VARCHAR(20)	NOT NULL	ID of the media that was copied

Book

Attribute Name	Data Type	Constraints	Description
ISBN	CHAR(13)	PK, 10-13 digits, unique	Identifying book number
Title	VARCHAR	NOT NULL	Book title
Author	VARCHAR	NOT NULL	Name of author
Publication Year	YEAR	NOT NULL	Year book was published

Genre	VARCHAR	NOT NULL	Type of genre
Count	INT	NOT NULL	How many on hand

Digital Media

Attribute Name	Data Type	Constraints	Description
ISBN	VARCHAR(20)	PK , NOT NULL	Unique identifier for digital media
Title	VARCHAR(100)	NOT NULL	Title of the digital media
Format	VARCHAR(50)	NOT NULL	Format type (DVD, eBook)
Creator	VARCHAR(50)	Maybe NULL	The main creator (author, director, etc.)
Release Year	YEAR	Maybe NULL	The year it was released
Genre	VARCHAR(50)	Maybe NULL	Genre or category
Count	INT	CHECK(	Number of copies available

Magazine

Attribute Name	Data Type	Constraints	Description
ID	VARCHAR(20)	PK , NOT NULL, UNIQUE	Unique identifier for the magazine (could be ISSN)
Title	VARCHAR(100)	NOT NULL	Title
Issue	VARCHAR(20)	NOT NULL	Issue number
Publication Date	DATE	Maybe NULL	Date the issue
Count	INT	CHECK(Count >= 0)	Number of copies of this issue

Transaction

Attribute Name	Data Type	Constraints	Description
<u>Transaction ID</u>	INT(12)	PK , NOT NULL	ID of the transaction
Checkout Date	DATE	NOT NULL	Date the item was checked out
Due Date	DATE	NOT NULL	Date the item is due
Return Date	DATE	Maybe NULL	Date the item was returned
Copy ID	INT(12)	NOT NULL	ID of the item being borrowed
Copy Type	VARCHAR(10)	NOT NULL	Type of media i.e. book or magazine

Reports

Attribute Name	Data Type	Constraints	Description
<u>Report ID</u>	INT(12)	PK , NOT NULL	ID of the report
Date of Generation	DATE	NOT NULL	Date the report was generated
Type	VARCHAR(...)	NOT NULL	Type of report i.e. borrow or return

Fine

Attribute Name	Data Type	Constraints	Description
<u>Fine ID</u>	INT(12)	PK , NOT NULL	ID of the fine
Copy ID	INT(12)	FK→Copy(CopyID)	ID of the related copy
Client ID	INT(12)	FK→Client(ClientID)	ID of the related client
Due Date	DATE	NOT NULL	Date the fine is due
Fine Amount	DECIMAL	NOT NULL	Amount due
Date Paid	DATE	Maybe NULL	Date the fine is paid

Membership

Attribute Name	Data Type	Constraints	Description
<u>MembershipTypeID</u>	INT(12)	PK , NOT NULL	Type of membership they have
TypeName	VARCHAR(...)	NOT NULL	Corresponding name for the TypeID
BorrowLimit	INT	NOT NULL	Maximum amount of books allowed to borrow
DailyFineRate	DECIMAL	NOT NULL	Cost per day kept past the due date

Reservation

Attribute Name	Data Type	Constraints	Description
<u>Reservation ID</u>	INT(12)	PK , NOT NULL	ID of the reservation
Client ID	INT(12)	FK→Client(ClientID)	ID of the client reserving the item
Copy ID	INT(12)	FK→Item(ItemID)	ID of the item reserved
Reservation Date	DATE	NOT NULL	Date the item was reserved
Status	VARCHAR(...)	NOT NULL	Status of the reservation (Reserved/Claimed)

Relationships

Generate

Attribute Name	Data Type	Constraints	Description
Report ID	INT(12)	PK , NOT NULL	ID of the report
Staff ID	INT(12)	NOT NULL	ID of staff making report

Staff_Return

Attribute Name	Data Type	Constraints	Description
Copy ID	INT(12)	NOT NULL	ID of specific copy
Staff ID	INT(12)	NOT NULL	ID of staff who returned book
Return Date	DATE	NOT NULL	Date copy was returned

Staff_Checks out

Attribute Name	Data Type	Constraints	Description
Copy ID	INT(12)	N	ID of specific copy
Staff ID	INT(12)		ID of staff who check out book
Checkout Date	DATE	NOT NULL	Date copy was checked out by staff

Add new

Attribute Name	Data Type	Constraints	Description
Copy ID	INT(12)	PK , NOT NULL	ID of copy being added
Type	VARCHAR(20)	NOT NULL	Type of media of copy
Staff ID	INT(12)	NOT NULL	ID of staff adding

Manage

Attribute Name	Data Type	Constraints	Description
Staff ID	INT(12)	NOT NULL	ID of staff managing client
Client ID	INT(12)	PK,NOT NULL	ID of client

Client_Return

Attribute Name	Data Type	Constraints	Description
Copy ID	INT(12)	NOT NULL	ID of copy being returned
Client ID	INT(12)	NOT NULL	ID of client returning
Return Date	DATE	NOT NULL	Date copy was returned

Reserves

Attribute Name	Data Type	Constraints	Description
Reservation ID	INT(12)	NOT NULL	ID of the reservation that is being made
Copy ID	INT(12)	NOT NULL	ID of the copy being reserved
Client ID	INT(12)	NOT NULL	ID of the person reserving the copy

Checks out

Attribute Name	Data Type	Constraints	Description
TransactionID	INT(12)	NOT NULL	ID of the transaction being performed
CopyID	INT(12)	NOT NULL	Item the transaction is being performed on

Makes

Attribute Name	Data Type	Constraints	Description
ClientID	INT(12)	FK, NOT NULL	ID of the person making the reservation
ItemID	INT(12)	FK, NOT NULL	ID of the item being reserved

Creates

Attribute Name	Data Type	Constraints	Description
ClientID	INT(12)	NOT NULL	ID of the person checking it out
CopyID	INT(12)	FK, NOT NULL	ID of the copy being checked out
TransactionID	INT(12)	FK, NOT NULL	ID of transaction being created

Generates

Attribute Name	Data Type	Constraints	Description
Staff ID	INT(12)	FK, NOT NULL	Who is making the report
Type	VARCHAR(10)	NOT NULL	What type of report is it

Pays

Attribute Name	Data Type	Constraints	Description
ClientID	INT(12)	FK, NOT NULL	ID of the client
FineID	INT(12)	FK, NOT NULL	ID of the fine

Have

Attribute Name	Data Type	Constraints	Description
<u>ClientID</u>	INT(12)	FK, NOT NULL	ID of the client

<u>MembershipTypeID</u>	INT(12)	FK, NOT NULL	Type of membership they have
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